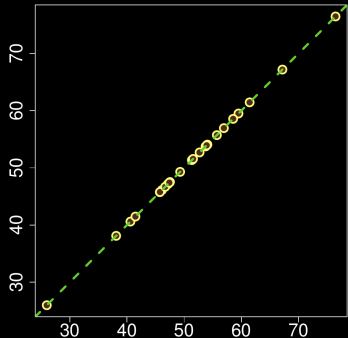


Chapter 8: Causality or relationship?

Throughout this course we've investigated various relationships.

There was evidence of a relationship between having a Canadian hometown and liking hockey.

There was evidence of a relationship between STAT 230 grade and STAT 231 grade.

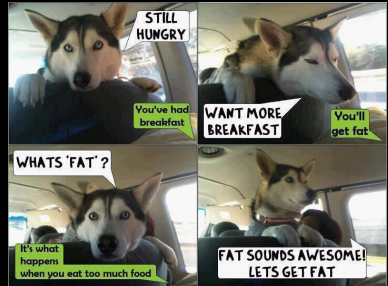


Relationship goals?

We've heard the phrase "correlation does not imply causation" - but what does 'causation' actually mean? 🤔

Causality or relationship?

Researchers have shown that there is a positive correlation between the average dietary fat intake and the breast cancer rate across countries. In other words, countries with higher intake tend to have higher breast cancer rates.



Does this prove that dietary fat **causes** breast cancer?

Studies have found that children with wealthier parents are more likely to go to college.

Does this mean that having wealthier parents **causes** people to go to college?

Parental Family Income and the Socioeconomic Attainment of Children

MARTHA S. HILL AND GREG J. DUNCAN

Survey Research Center, Institute for Social Research, University of Michigan

Like others before us using different data, we find significant effects of parental family income on the completed schooling and wage rates of adult children using intergenerational data from the Panel Study of Income Dynamics. We explore

What does the statement “X causes Y” mean?

Does smoking cause lung cancer? 🤔

Does a high STAT 230 grade cause a high STAT 231 grade? 🤔

Does the location of your hometown cause whether you like hockey? 🤔

Gravity causes dropped objects to fall to the ground.

Does smoking cause lung cancer?

Does everyone who smokes get lung cancer?

In most cases causation is difficult to define.



BRITISH MEDICAL JOURNAL

LONDON SATURDAY JUNE 26 1954

THE MORTALITY OF DOCTORS IN RELATION TO THEIR SMOKING HABITS

A PRELIMINARY REPORT

BY

RICHARD DOLL, M.D., M.R.C.P.

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In the last five years a number of studies have been made of the smoking habits of patients with and without lung cancer (Doll and Hill, 1950, 1952; Levin, Goldstein, and Gerhardt, 1950; Mills and Porter, 1950; Schrek, Baker, Ballard, and Dolgoff, 1950; Wynder and Graham, 1950; McConnell, Gordon, and Jones, 1952; Koulumies, 1953; Sadowsky, Gilliam, and Cornfield, 1953; Wynder and Cornfield, 1953; Breslow, Hoaglin, Rasmussen, and Abrams, 1954; Watson and Conte, 1954). All these studies agree in

tionary. In addition to giving their name, address, and age, the doctors were asked to classify themselves into one of three groups—namely, (a) whether they were, at that time, smoking; (b) whether they had smoked but had given up; or (c) whether they had never smoked regularly (that is, had never smoked as much as one cigarette a day, or its equivalent in pipe tobacco, for as long as one year). All present smokers and ex-smokers were asked additional questions. The former were asked the ages at which they had started smoking

TABLE IV.—Standardized Death Rate Per Annum Per 1,000 Men Aged 35 Years and Above in Relation to the Most Recent Amount of Tobacco Smoked

Cause of Death	No. of Deaths Recorded	Death Rates of Non-smokers	Death Rates of Men Smoking a Daily Average of			Death Rate of All Men
			1 g.—	15 g.—	25 g.+	
Lung cancer	36*	0.00	0.48	0.67	1.14	0.66
Other cancers	92	2.32	1.41	1.50	1.91	1.65
Respiratory disease (other than cancer)	54	0.86	0.88	1.01	0.77	0.94
Coronary thrombosis	235	3.89	3.91	4.71	5.15	4.27
Other cardiovascular diseases	126*	2.23	2.07	1.58	2.78	2.14
Other diseases	247	4.27	4.67	3.91	4.52	4.36
All causes	789	13.61	13.42	13.38	16.30	14.00

* 1 case in which lung cancer was recorded as a contributory but not a direct cause of death has been entered in both groups.

55-64 when the questionnaire was completed subsequently died of lung cancer. The proportions of all the men in this age group who were non-smokers, smoked 1-14 g. a day, 15-24 g. a day, or 25 or more g. a day were 9.3%, 33.9%, 31.6%, and 25.3%. If the mortality from lung cancer is unrelated to smoking, then the 13 deaths should be distributed to the smoking groups in these ratios. Similar calculations have been made for the numbers of men dying of lung cancer in the other age groups—namely, 1 at ages 35-44, 8 at ages 45-54, 6 at ages 65-74, and 8 at ages 75-84. The total numbers of deaths expected in each smoking category were then obtained by adding the numbers for the separate age groups. The results were as follows:

	Non-smokers	Smokers of a daily average of			Total
		1 g.—	15 g.—	25 g.+	
Observed deaths	0	12	11	13	36
Expected deaths	3.77	14.20	10.73	7.33	36.03

These differences between the observed and expected deaths are statistically significant ($\chi^2=8.5$, $n=3$, $P=0.04$). We may

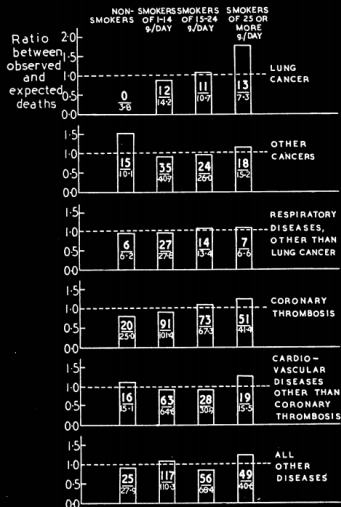
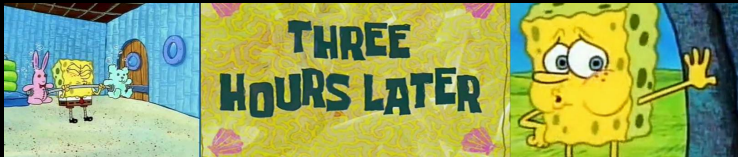


Chart showing variation in mortality with amount smoked. The ordinate shows the ratio between the number of deaths observed and the number expected (as entered in each column).

Consider the following definition:

Let y be a response variate and let x be an explanatory variate associated with units in a population or process. Then, if all other factors that affect y are held constant, let us change x (or observe different values of x) and see if y changes. If it does we say that x has a causal effect on y .



If x = how much exercise you've done, and y = how out of breath you feel, then an increase in x might cause an increase in y .

Suppose we are interested in whether submitting STAT 231 assignments for feedback increases the probability a student passes the course.

We could compare the grades of students who submit their assignments for feedback with those who don't.

Let's think about this in the context of the preceding definition of causality.

Question: the decision to submit an assignment is voluntary. What implications might this have for establishing causal relationships?



Let y be a response variate and let x be an explanatory variate associated with units in a population or process. Then, if all other factors that affect y are held constant, let us change x (or observe different values of x) and see if y changes. If it does we say that x has a causal effect on y .

In many situations, we may find differences between groups that do/do not take some kind of action (submit an assignment, take a drug, etc.).

This is why 'if all other factors that affect y are held constant' is an important part of establishing causality.

Suppose we compare students who have submitted at least one assignment with those who have not:

- x denote whether someone has submitted at least one assignment ($x = 1$) or not ($x = 0$); and
- y denote whether or not the student passes the course ($y = 1$) or not ($y = 0$).

Our preceding definition of causality suggests we should see if y changes when x changes.

Many students will pass the course without submitting assignments.

Some students might fail the course despite submitting assignments.

We are more interested in whether the *probability* of passing is higher among those who submit assignments.

The preceding definition is not broad enough since a change in x may only lead to a change in the *distribution* of y .

Improved Definition: x has a causal effect on Y if, when all other factors that affect Y are held constant, a change in x induces a change in a property of the distribution of Y .

For example, let Y = the number of students in a random sample of n students who pass the course. Assume $Y \sim \text{Binomial}(n, \theta)$.

We could say that submitting assignments has a causal effect on passing the course if the value of θ was different for people who submitted at least one assignment and those who had not, assuming everything else that affects Y is held constant.

Improved Definition: x has a causal effect on Y if, when all other factors that affect Y are held constant, a change in x induces a change in a property of the distribution of Y .

Unfortunately this definition is impractical since we cannot hold all other factors that affect y constant. (We may not even know what all the factors are!)

The definition serves as an ideal that should be used to conduct studies in order to show that a causal relationship exists.

Reasons two variates can be related

We've already talked about some reasons why there may be an association between two variates. We'll now take a more methodical approach and consider six possible explanations:

- 1 The explanatory variate is the direct cause of the response variate.
- 2 The response variate is causing a change in the explanatory variate.
- 3 The explanatory variate is a contributing but not sole cause of the response variate.
- 4 Both variates are changing with time.
- 5 The association may be due to coincidence.
- 6 Both variates may result from a common cause.



Reason 1: The explanatory variate is the direct cause of the response variate.

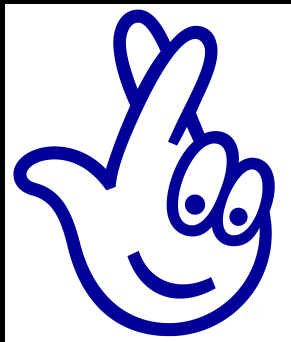
Occasionally, a change in the explanatory variate is the direct cause of a change in the response variate. For example, if I'm thirsty, drink a cup of tea, and am then not thirsty, we would probably agree that the act of drinking tea caused my thirst to decrease.



Unfortunately, even if one variate is the direct cause of another, we may not see a strong association.

For example, playing the lottery is the direct cause of winning the lottery.

But the relationship between playing the lottery and winning the lottery is not strong; most decisions to play the lottery do not result in winning it.



Reason 2: The response variate is the direct cause of the explanatory variate.

Example: Many video game players engage in what is known as 'toxic' behaviour, such as insulting opponents, or engaging in unsporting conduct in the game.

Suppose a study was designed to investigate the effect of 'toxic' behaviour of video game players on their performance in games. We might suppose:

- Explanatory variate = gamer toxicity level.
- Response variate = proportion of matches won.

We might expect toxic behaviour to cause players to lose more often, as engaging in the behaviour may distract or discourage their teammates, or even cause the player to play worse.

Sometimes the causal connection is the opposite of what we might expect!

For example, perhaps players are more likely to engage in toxic behaviour as a result of losing matches.

This could be an example of 'reverse causality'.



Me when my teammates are trash at the game.



Me when I'm trash at the game.

Reason 2: The response variate is the direct cause of the explanatory variate.

Some studies have shown that people who drink diet (low sugar) sodas are more likely to be overweight.

Does this suggest drinking diet soda causes people to become overweight?



British Pepsi Max is best
Pepsi Max don't @ me.

Reason 3: The explanatory variate is a contributing, but not the only, cause of the response variate.

The kinds of complex phenomena most often studied by researchers are likely to have multiple causes.

Even if there was a causal connection between diet and a type of cancer, for instance, it would be highly unlikely that the cancer was caused solely by eating that certain type of diet.



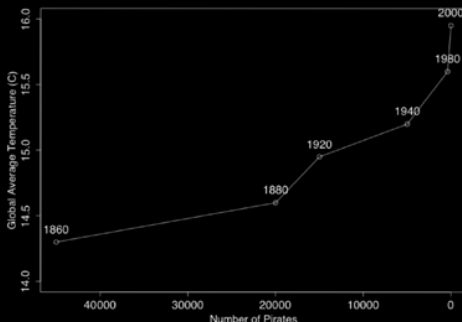
It is particularly easy to be misled into thinking you have found a sole cause for a particular outcome, when you have actually found a *necessary* contributor to the outcome.

For example, the presence of a specific gene mutation is required in order for certain cancers to occur. However, presence of the mutation alone does not guarantee someone will develop that form of cancer.

Reason 4: Both variates are changing over time.

Nonsensical associations often result from correlating two variates that are both changing over time.

For example, there is a strong (negative) correlation between global average temperature, and the number of pirates.



Source: Henderson, R. 2006, *The Gospel of the Flying Spaghetti Monster*, Villard.

Reason 5: The association may be nothing more than coincidence, or random chance.

Example: Thomas Jefferson, third president of the United States, died shortly after noon on July 4, 1826. A few hours later, John Adams, second president of the United States, also died.

Would we conclude that Jefferson's death (which occurred first) caused the death of Adams?

Example: Suppose a new office building opened and within a year there was an unusually high rate of brain cancer among workers in the building.

Suppose someone calculated that the odds of having that many cases in one building were only 1 in 10,000.

We might suspect that something wrong in the environment was causing people to develop brain cancer.



**TURN OFF
CELL PHONES**

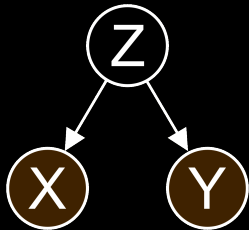
The problem with this reasoning is that it focuses on the odds of seeing such a rare event in that particular building in that particular city. It fails to take into account that there are thousands of new office buildings.

If the odds really were only 1 in 10,000, we should expect to see this phenomenon just by chance in about 1 of every 10,000 buildings.

It would be unusual if we did not occasionally see clusters of diseases as chance occurrences.

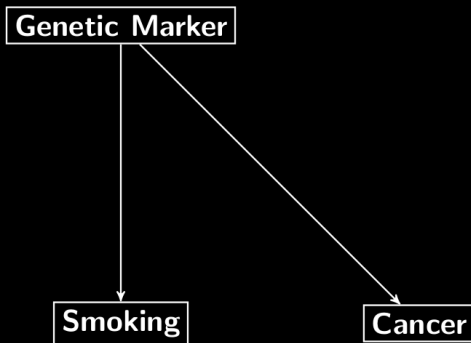
Reason 6: Both variates may result from a common cause.

An association between two variates may be observed because both variates are responding to changes in some unobserved variate or variates.

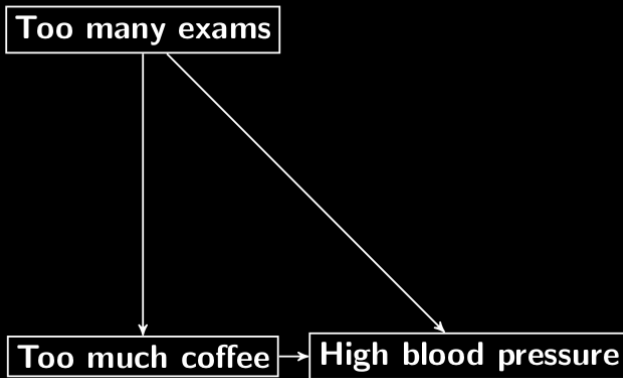


These variates are sometimes referred to as confounding or lurking variates.

For example, a much used explanation by the tobacco companies for the association between smoking and lung cancer is that smoking behaviour and lung cancer are both responses to a genetic predisposition.



As another example, suppose hot chocolate and sweater sales increase at the same time. We would not conclude that hot chocolate sales *cause* sweater sales.



What is the confounding variate here? 🤔

Lurking variates can lead to seemingly counterintuitive results.

A study into two treatments for kidney stones involved 700 patients, with the following results:

- Treatment A: 273/350 successful treatments, or 78%
- Treatment B: 289/350 successful treatments, or 83%.

However, treatments weren't assigned at random (more on this in a moment)! If the kidney stones are categorized as 'small' or 'large', we see the following:

	Treatment A	Treatment B
Small	93% (81/87)	87% (234/270)
Large	73% (192/263)	69% (55/80)
Both	78% (273/350)	83% (289/350)

Treatment A is better in both groups!

	Treatment A	Treatment B
Small	93% (81/87)	87% (234/270)
Large	73% (192/263)	69% (55/80)
Both	78% (273/350)	83% (289/350)

This is an example of what is known as *Simpson's Paradox*.

The reason for the apparent paradox is that the size of the kidney stone is a confounding or lurking variate.

Simpson's Paradox shows up in lots of places, and can be difficult to spot!

Two particularly common causes of Simpson's Paradox are:

- Symptom severity (e.g., if patients with more severe symptoms are given a particular treatment).
- Age (e.g., if there are variations in health outcomes for older/younger people).

Way back in Lecture 1 we saw this headline from the New York Times:

The New York Times

Covid and Race

The death rate for white Americans has recently exceeded the rates for Black, Latino and Asian Americans.

Simpson's Paradox



Tyler Black, MD ✓

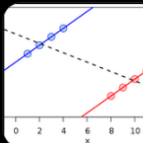
@tylerblack32

...

The entire article is based upon a mathematical fallacy known as Simpson's paradox.

Rudimentary statistics.

An embarrassing mistake by an influential paper and author that distorts the impact of racial discrimination.



en.wikipedia.org

Simpson's paradox - Wikipedia

Simpson's Paradox



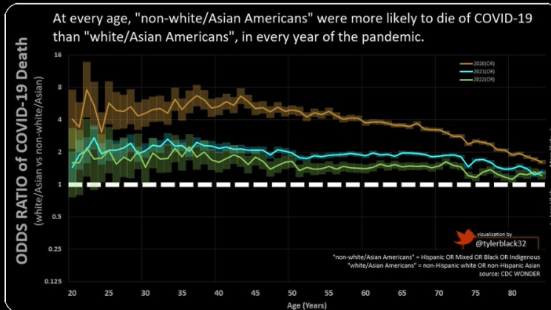
Tyler Black, MD
@tylerblack32

...

I did the math that @DLeonhardt and the @nytimes couldn't do, because, unlike him, I know how to use mortality statistics.

Age standardization matters.

Graph: On all three years of the pandemic, non-white/Asian Americans had a higher odds ratio of #COVID death at every age.



Reasons two variates can be related

We've now seen six different explanations for why two variates might be associated:

- 1 The explanatory variate is the direct cause of the response variate.
- 2 The response variate is causing a change in the explanatory variate.
- 3 The explanatory variate is a contributing but not sole cause of the response variate.
- 4 Both variates are changing with time.
- 5 The association may be due to coincidence.
- 6 Both variates may result from a common cause.

Establishing cause and effect

Given the number of possible explanations for association between two variates, how do we ever establish that there is actually a causal connection?

Ideally, we would find groups of people who are - on average - the same, give members of each group a different treatment, then compare the response measures in each group.

This would account for the problem of confounding variates.



Establishing cause and effect

How can we find similar groups of people?

We could use groups of identical twins, but this is rarely practical.



One powerful technique: **randomization**.

In the kidney stone example, small kidney stones were easier to treat, and Treatment B was mostly used on patients with small kidney stones.

	Treatment A	Treatment B
Small	93% (81/87)	87% (234/270)
Large	73% (192/263)	69% (55/80)
Both	78% (273/350)	83% (289/350)

If treatments had been randomized we might expect a more even distribution of treatments across the two sizes of kidney stones.

Smoking and lung cancer

Randomization is, however, not always possible.

For example, how could we use randomization to establish a link between smoking and lung cancer?



Another example: 8.2.2 in Course Notes - designing a good experiment can be surprisingly complicated!

Establishing causation in observational studies

In observational studies controlling variates and using randomization is not possible. Establishing causation in observational studies is much more difficult and requires at least the following four features.

We can see that each appears to be satisfied in the case of smoking and lung cancer.

1) The association between the two variates must be observed in many studies of different types among different groups. This reduces the chance that an observed association is due to a defect in one type of study or a peculiarity in one group of subjects.

A strong association between smoking and lung cancer has been observed in numerous studies in many countries.

Establishing causation in observational studies

2) The association must continue to hold when the effects of plausible confounding variates are taken into account.

Many possible sources of confounding variates have been examined in these studies and have not been found to explain the association.

For example, data about nonsmokers who are exposed to secondhand smoke contradicts the genetic hypothesis.

Establishing causation in observational studies

3) There must be a plausible scientific explanation for the direct influence of one variate on the other variate, so that a causal link does not depend on the observed association alone.

Animal experiments have demonstrated conclusively that tobacco smoke contains substances that cause cancerous tumors.

Therefore there is a known pathway by which smoking causes lung cancer.

Establishing causation in observational studies

4) There must be a consistent response, that is, one variate always increases (decreases) as the other variate increases.

The lung cancer rates for ex-smokers decrease over time since smoking cessation.

The evidence for causation here is about as strong as non-experimental evidence can be.

The *Bradford Hill criteria* are a (slightly outdated) set of guidelines similar to these, which make for interesting reading (but are not part of STAT 231).



In our final chapter, we have explored the concept of causality, from how it's defined, to reasons why a causal relationship may (or may not) exist between our variates. Key concepts:

- Defining causality: We look for a change in the *distribution* of a response variate, not simply a change in its observed value.
- Reasons for association: We outlined six reasons why two variates may be associated.
- Randomization: One way to establish causality from observational data.

PROTIP: Try applying these ideas to things you see in everyday life, such as news stories or other events.

Chapter 8 gives a small snapshot into what is a huge area of statistics known as causal inference.

If you're interested in learning more about this topic beyond what's covered in STAT 231, please ask!

But on that note, we've reached the end of the course. 🥲

“The statistician who supposes that their main contribution to the planning of an experiment will involve statistical theory, finds repeatedly that they make their most valuable contribution simply by persuading the investigator to explain why they wish to do the experiment.” - Gertrude Mary Cox

